

Model Question paper

QP CODE:

PAGES:3

Reg No:_____

Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SEVENTH SEMESTER B.TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: CXT401

Course Name: User Interface Software & Technology

Max.Marks:100

Duration: 3 Hours

PART A

Answer all Questions. Each question carries 3 Marks

1. Define user research in the context of interaction design. Provide two examples of user research methods commonly used in interface design.
2. Explain the concept of "satisficing" in user behavior. How can designers address satisficing tendencies in interface design?
3. Discuss the significance of information architecture in interface design. Provide two principles of effective information architecture.
4. How can lists be used to organize contents in Bootstrap5?
5. Explain the importance of clear entry points in interface design. Provide two examples of interfaces that effectively utilize clear entry points.
6. Explain navbar with its subcomponents.
7. Discuss about grid of equals in page layout design.
8. Explain the concept of module tabs in page layout. How do module tabs help organize content on a webpage?
9. Explain the concept of "responsive design" in interface development. How does responsive design enhance user experience across different devices?
10. Explain form control in Bootstrap5.

Part B

(Answer any one question from each module. Each question carries 14 Marks)

11. a) Discuss the significance of user research in interaction design, considering its impact on product development and user satisfaction. Provide examples to illustrate the importance of user research in real-world interface design projects. (7 marks)
- b) Explain the concept of "instant gratification" in user behavior and its implications for interface design. How can designers balance the need for instant gratification with other design considerations such as usability and accessibility? (7 marks)

OR

12. a) Discuss the concept of "streamlined repetition" in interface design and its relationship to user efficiency and productivity. How can designers leverage streamlined repetition to improve user workflows and task completion rates? (7 marks)
- b) Assess the role of habituation in user behavior and its effects on interface engagement over time. Provide strategies for designing interfaces that mitigate habituation and maintain user engagement (7 marks)
13. a) Explain the effectiveness of different information architecture models in organizing complex content structures. Provide examples of interfaces that demonstrate the successful implementation of each model. (7 marks)
- b) Explain how the component Table is used to organize content in Bootstrap5. (7 marks)

OR

14. a) Discuss the role of alternative views in content organization and navigation design. How do alternative views enhance user comprehension and exploration of diverse content types? (7 marks)
- b) Explain the effectiveness of the canvas plus palette pattern in organizing and presenting content within digital interfaces. (7 marks)
15. a) How do different navigational models influence user comprehension and task completion rates? (7 marks)
- b) Discuss the advantages and limitations of modal panels and provide recommendations for their appropriate use in interface design. (7 marks)

OR

16. a) Discuss the concept of "escape hatch" in navigation design and its role in facilitating user exploration and recovery from navigational errors. (7 marks)
- b) Explain the effectiveness of sign-in tools in interface navigation design, considering their impact on user authentication processes and overall user experience. (7 marks)
17. a) Explain the principles of visual hierarchy in page layout design and their influence on user

attention and comprehension.

(7 marks)

b) Discuss the best practices for implementing accordion panels in interface design.

(7 marks)

OR

18. a) How can designers leverage liquid layouts to optimize user experience across different devices and screen sizes? (7 marks)

b) Explain the effectiveness of diagonal balance in page layout design, considering its impact on visual aesthetics and user engagement. (7 marks)

19. a) How can designers implement progressive disclosure techniques to enhance user understanding and engagement? (7 marks)

b) Discuss the best practices for designing intuitive and accessible hover interactions. (7 marks)

OR

20. a) How can designers implement multi-level undo functionality effectively while maintaining interface consistency and performance? (7 marks)

b) Discuss the best practices for designing and implementing macros in interfaces to support user workflows and task automation. (7 marks)